A GEOMETRIC FIBRE OF A MAP OF SPACES IS CONNECTED EXACTLY WHEN EVERY ETALE NEIGHBOURHOOD SEES A CONNECTED ORDINARY FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8098-8105 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8107-8124, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side of a pair restatement; schemes twin uncounted per ruling.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Properties of Spaces, lemma cofinal-etale; Spaces over Fields, section geometrically-connected and lemma characterize-geometrically-disconnected; More on Morphisms, lemma realize-prescribed-residue-field-extension-etale.

Source reference. Original lemma 02LF: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8124–8124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A17: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8124–8124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A0Y: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8124–8124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04JW: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8124–8124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\overline{y}$ be a geometric point of Y. Then $X_{\overline{y}}$ is connected, if and only if for every étale neighbourhood $(V, \overline{v}) \to (Y, \overline{y})$ where V is a scheme the base change $X_V \to V$ has connected fibre X_v.*

Proof. Since the category of étale neighbourhoods of $\overline{y}$ is cofiltered and contains a cofinal collection of schemes (Properties of Spaces, Lemma 04JW) we may replace Y by one of these neighbourhoods and assume that Y is a scheme. Let $y \in Y$ be the point corresponding to $\overline{y}$. Then X_y is geometrically connected over $\kappa(y)$ if and only if $X_{\overline{y}}$ is connected and if and only if $(X_y)_{k'}$ is connected for every finite separable extension k' of $\kappa(y)$. See Spaces over Fields, Section 0A0Y and especially Lemma 0A17. By More on Morphisms, Lemma 02LF there exists an affine étale neighbourhood $(V, v) \to (Y, y)$ such that $\kappa(s) \subset \kappa(u)$ is identified with $\kappa(s) \subset k'$ any given finite separable extension. The lemma follows. □